

Problem Statement

Inefficiencies in management of classroom spaces.

- Financial and energy waste
- Sub-optimal learning environments
- Slow strategic planning
- Security and safety concerns



Project Objective

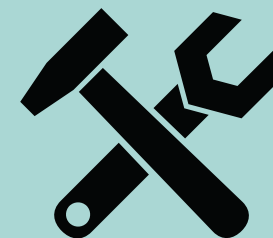
Develop a tool that uses AWS that help a university manage their resources



Optimise space utilisation



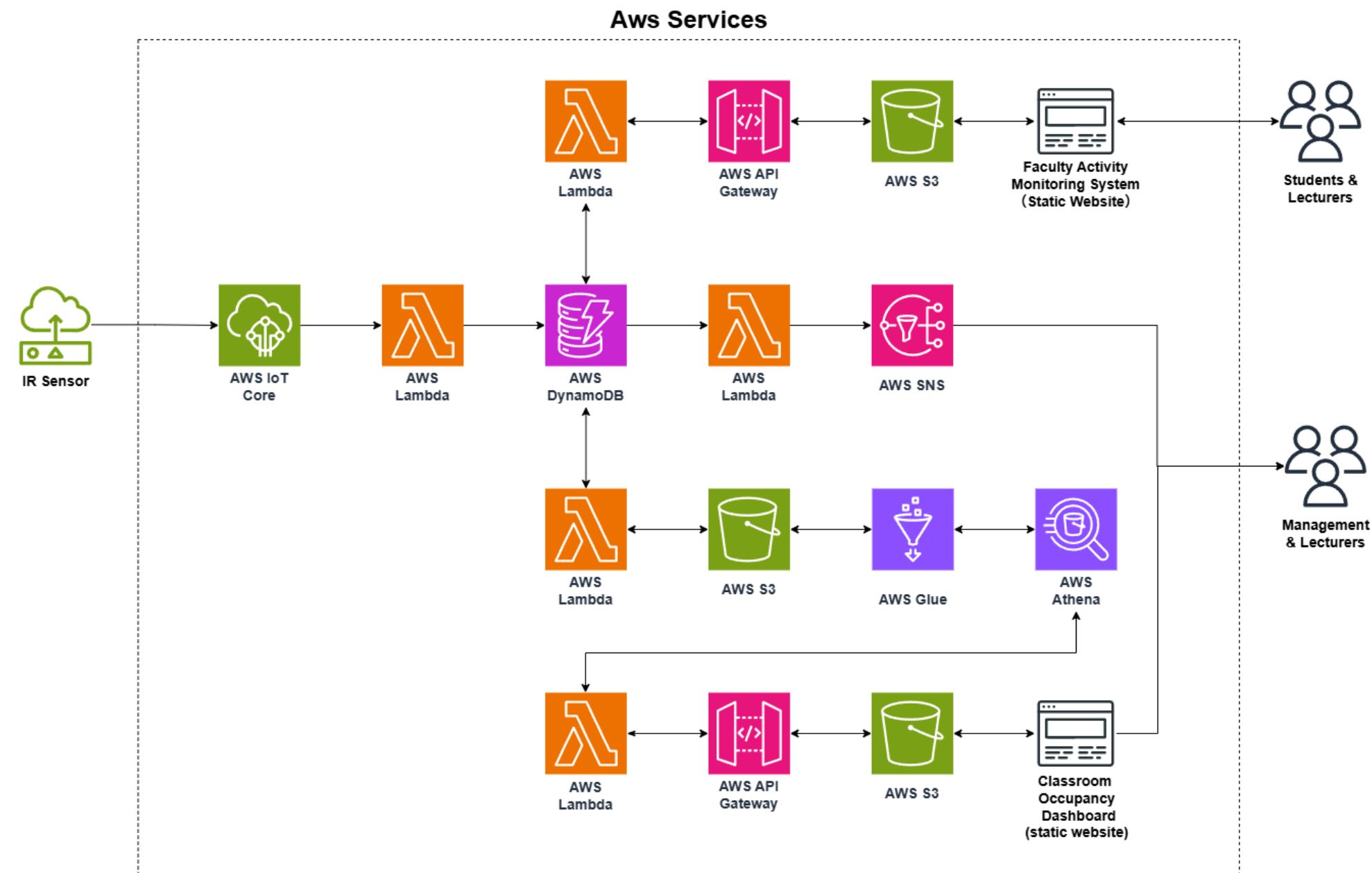
Make Informed Decisions



Provide real-time or manual adjustments

System Architecture

Faculty Activity Monitoring & Space Utilisation Analytics



IoT Core

- Connect and Manage IoT devices
- Data collection and routing
- Secure communication



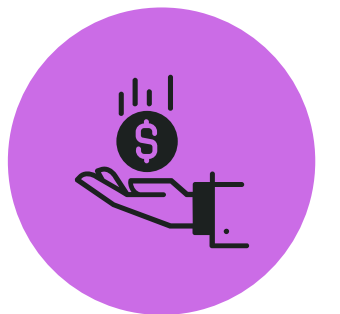
Lambda

- Process incoming data
- Execute custom logic
- Triggers AWS DynamoDB



DynamoDB

- Fast, scalable NoSQL database
- Store IoT data
- Allow fast access and queries



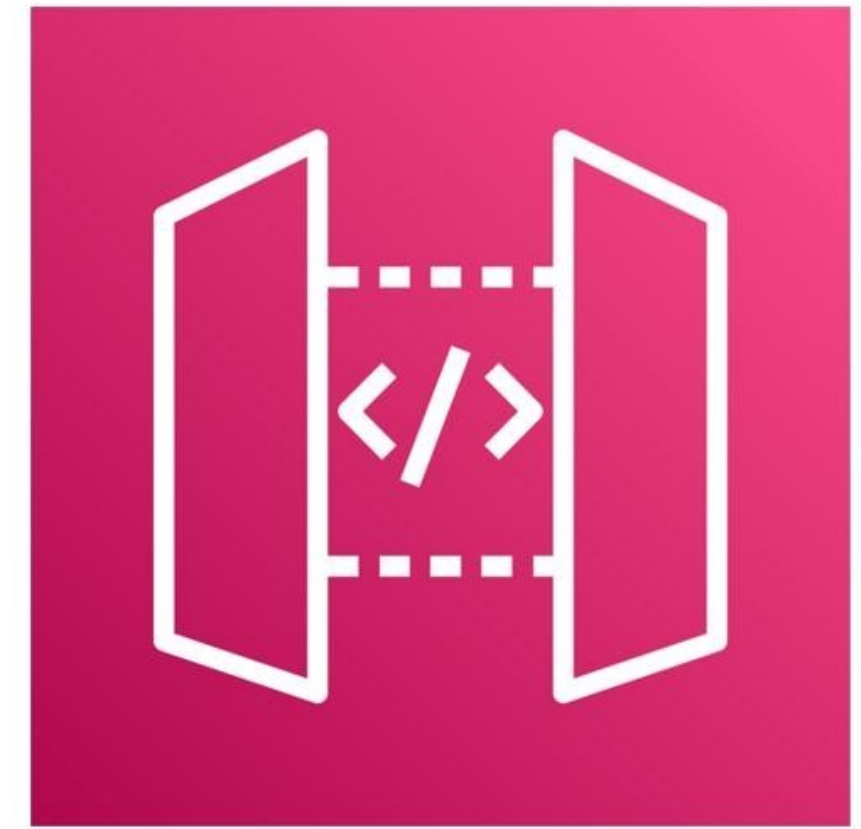
Lambda

- Executes backend code in response to an API Gateway request.
- Read data from DynamoDB.
- Sends the response back to API Gateway.



API

- Exposes HTTP endpoints
- Acts as a gateway between clients and backend services.
- Receives API requests and forwards them to Lambda.



S3

- Hosts static files like HTML, CSS, JavaScript, and images.
- Serves as the front-end.



Lambda

- Fetch data from DynamoDB
- Put them into S3



S3

- Serves as a data lake
- Keeps track of every changes that has been made to the database



Glue

- Used to crawl data from the data lake
- Format the data into a relational database format



Athena

- Perform SQL queries on the database created from glue



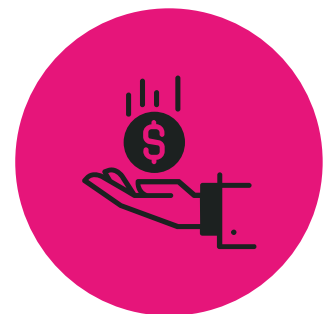
Lambda

- Fetch data from DynamoDB
- Trigger SNS



SNS

- Receives alert messages from the Lambda function.
- Distributes alerts to all subscribers (Email, etc.).



Thank you very much!

PRESENTED BY GROUP 13